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Reaction 1

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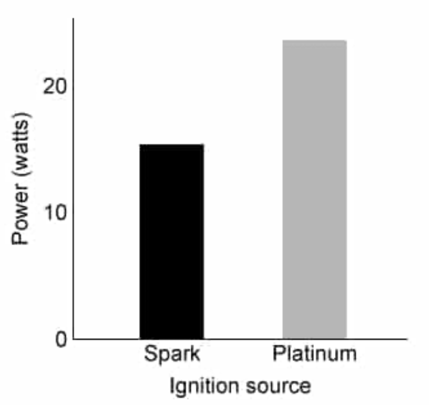

Figure 1 Relative power outputs of hydrogen engines with different ignition sources at 298 K

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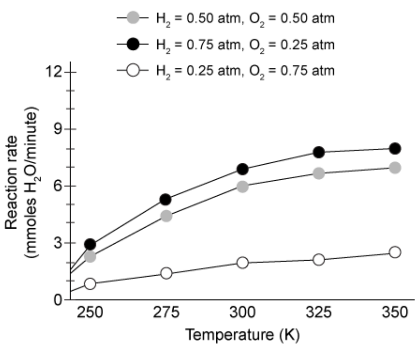


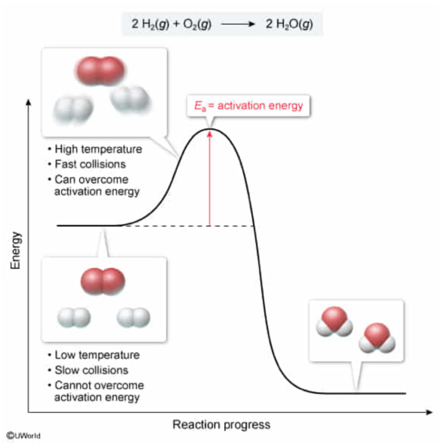
Figure 2 Rate of combustion in platinum-coated engines under various conditions

In the absence of platinum, a spark initiates combustion by doing which of the following?

- A. It shifts the reaction equilibrium toward water formation. [17%]
- B. It consumes oxygen, driving the reaction toward the products. [18%]
- ✔ C. It increases the average kinetic energy of nearby molecules. [57%]
- D. It removes electrons from oxygen to form reactive oxygen species. [7%]

Correct
56% answered correctly

Explanation:



Most chemical reactions can occur only when two or more molecules collide with enough energy to break existing bonds so that new bonds can be formed. This minimum amount of energy required to initiate the reaction is called the activation energy. If the resulting reaction is exothermic (ie, more energy is released than is consumed to initiate the reaction), the released energy sustains the reaction by enabling other molecules to react.

Temperature is a measure of the average kinetic energy of the molecules in a system. At higher temperatures, the molecules move more quickly and collide more often and with more energy. As such, an increase in temperature **increases the rate of a reaction** by providing more molecules with the required activation energy.

The reaction between hydrogen and oxygen is thermodynamically favorable (ie, can occur spontaneously); however, the reaction has a high activation energy. At room temperature, most of the molecules lack the energy necessary to initiate the reaction, and the reaction proceeds so slowly that it effectively does not occur at all. A spark releases a small amount of heat into the system, raising the localized temperature and kinetic energy of nearby molecules surrounding the spark. These molecules are then able to react exothermically, releasing more heat into the system and providing more molecules with enough energy to react.

(Choice A) The equilibrium position of a reaction is not appreciably altered by the presence of a spark. Instead, the rate at which equilibrium is achieved changes.

(Choice B) The spark consumes a negligible amount of oxygen. Because oxygen is a reactant, its consumption would drive the reaction toward reactant formation based on Le Châtelier's principle.

(Choice D) Oxidizers such as oxygen gain electrons in combustion reactions. They do not lose electrons.

Educational objective:
Molecules must collide with sufficient energy (known as the activation energy) for a reaction to occur. Increased temperature increases the kinetic energy of the molecules in a system, and therefore increases the rate of a reaction.

Time spent: 0 sec
Subject: General Chemistry

QID: 401200
System: Thermodynamics, Kinetics & Gas Laws

Combustion occurs when an oxidation-reduction reaction takes place between a reduced fuel source and an oxidizer, most frequently oxygen. The reaction between hydrogen and oxygen forms water (Reaction 1) and is highly exothermic and thermodynamically favorable. At the completion of hydrogen combustion, essentially all H_2 molecules have been converted to water. Nevertheless, at room temperature and atmospheric pressure, molecular hydrogen and oxygen can coexist quite stably.

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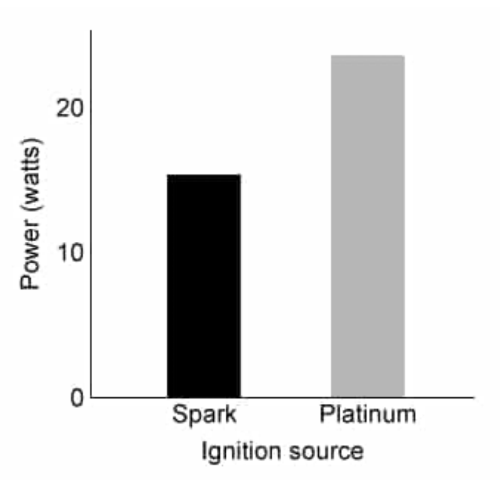

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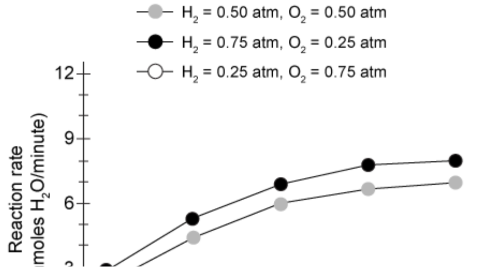


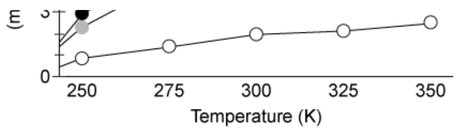


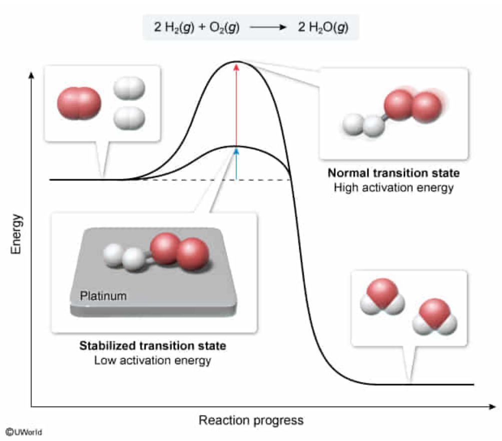
Figure 2 Rate of combustion in platinum-coated engines under various conditions

Platinum most likely increases the reaction rate by stabilizing:

- A. the reactants. [2%]
- B. the products. [3%]
- C. the intermediate step. [6%]
- ✔ D. the transition state. [88%]

Correct
88% answered correctly

Explanation:



The transition state of a reaction is the state of the molecules in which existing bonds are broken and new bonds are forming. It is an **unstable state** and normally requires a large amount of activation energy to form. As such, the activation energy acts as an **energy barrier** to a reaction's progress.

A **catalyst** is a substance that increases the rate of a reaction without being consumed by the reaction. Catalysts increase the rate by stabilizing the transition state and thereby **lowering the activation energy** of the reaction. The passage states that platinum increases the reaction rate of hydrogen combustion without being consumed; therefore, platinum must be acting as a catalyst, stabilizing the transition state of the combustion reaction.

(Choices A and B) Stabilization of reactants or products would not help a reaction proceed more quickly because the energy barrier between them (the transition state) would still be present.

(Choice C) An intermediate step is a relatively stable step in a reaction that exists between two transition states. Its stabilization would not help increase the rate of a reaction.

Educational objective:
The transition state of a reaction is unstable and presents an energy barrier to the reaction's progress. A catalyst stabilizes the transition state, and therefore increases the reaction rate without being consumed.

Time spent: 0 sec
Subject:
General Chemistry

QID: 401201
System:
Thermodynamics, Kinetics & Gas Laws

Combustion occurs when an oxidation-reduction reaction takes place between a reduced fuel source and an oxidizer, most frequently oxygen. The reaction between hydrogen and oxygen forms water (Reaction 1) and is highly exothermic and thermodynamically favorable. At the completion of hydrogen combustion, essentially all H_2 molecules have been converted to water. Nevertheless, at room temperature and atmospheric pressure, molecular hydrogen and oxygen can coexist quite stably.

$$2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(g)$$

Reaction 1

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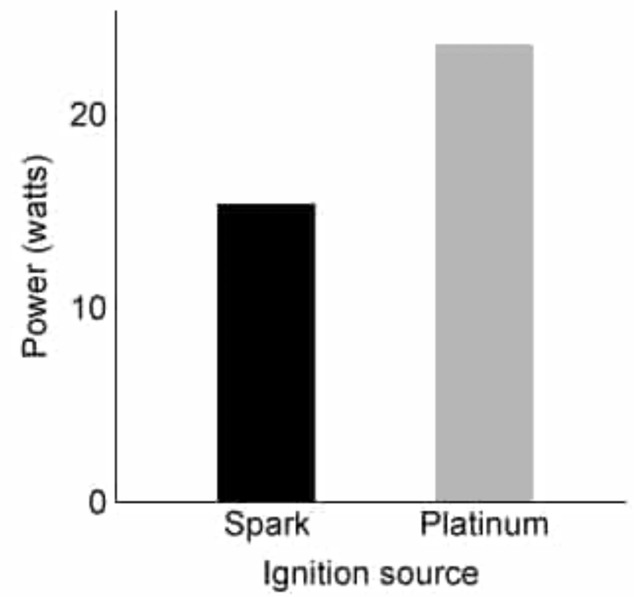

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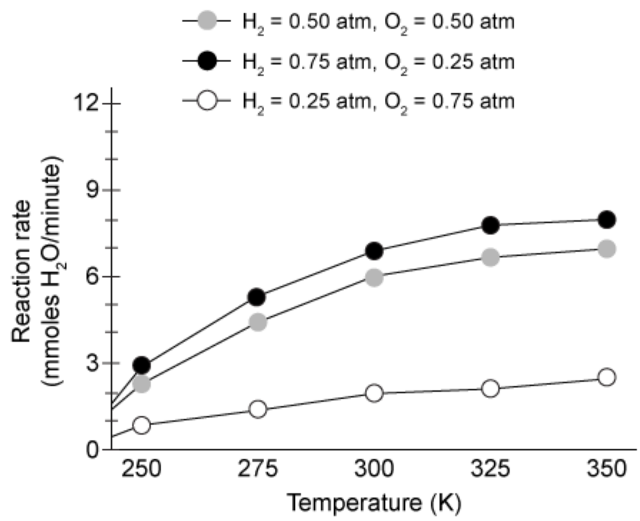

Figure 2 Rate of combustion in platinum-coated engines under various conditions

Assuming the combustion of hydrogen behaves like an elementary reaction with respect to its rate, which of the following explains why an increase in the H_2/O_2 ratio causes an increase in the reaction rate?

- A. Pauli exclusion principle [1%]
- B. Law of mass action [53%]
- C. Boyle's law [21%]
- D. Henry's law [23%]

Correct
52% answered correctly

Explanation:

Law of mass action applied to elementary reaction rates

$$2\,H_2 + O_2 \xrightarrow{k_1} 2\,H_2O$$

$$\text{Rate} = k_1(P_{H_2})^2(P_{O_2})$$

The **law of mass action** is usually expressed with respect to the equilibrium constant of a reaction. However, when applied to the dynamic equilibrium of an elementary reaction, the law of mass action also states that the **rate of a reaction** is proportional to the **molar amount** (ie, the molar concentration or partial pressure) of each reactant **raised to the power** of its reaction order. For an elementary reaction (ie, a reaction that proceeds in a single step), the reaction order of each species is equal to its stoichiometric coefficient. For non-elementary reactions, the order must be determined empirically.

Many non-elementary reactions behave as if they were elementary with respect to the reaction rate. Assuming the combustion of hydrogen is one such non-elementary reaction, the reaction is second order with respect to H_2 but only first order with respect to O_2. Therefore, the rate of the reaction depends on the partial pressure of O_2 and on the square of the partial pressure of H_2. As long as enough O_2 is provided for the reaction to proceed, removing some O_2 and replacing it with the same number of moles of H_2 will result in a rate increase because H_2 makes a greater contribution to the rate than O_2 does.

(Choice A) The Pauli exclusion principle states that no two electrons in an atom can have the same set of quantum numbers. It is not related to the reaction rate.

(Choice C) Boyle's law says that the pressure of a gas is inversely proportional to the volume it occupies. It does not directly predict the rate of any reaction.

(Choice D) Henry's law states that the amount of a gas that dissolves in a liquid is proportional to the partial pressure of that gas. It is not predictive of the reaction rate.

Educational objective:
The law of mass action states that the rate of a reaction is proportional to the molar amount of each reaction component raised to the power of its reaction order. For elementary reactions, the reaction order of each species is equal

to its stoichiometric coefficient.

Time spent:0 sec	**QID**:401202
Subject:	**System:**
General Chemistry	Thermodynamics, Kinetics & Gas Laws

Combustion occurs when an oxidation-reduction reaction takes place between a reduced fuel source and an oxidizer, most frequently oxygen. The reaction between hydrogen and oxygen forms water (Reaction 1) and is highly exothermic and thermodynamically favorable. At the completion of hydrogen combustion, essentially all H_2 molecules have been converted to water. Nevertheless, at room temperature and atmospheric pressure, molecular hydrogen and oxygen can coexist quite stably.

$$2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(g)$$

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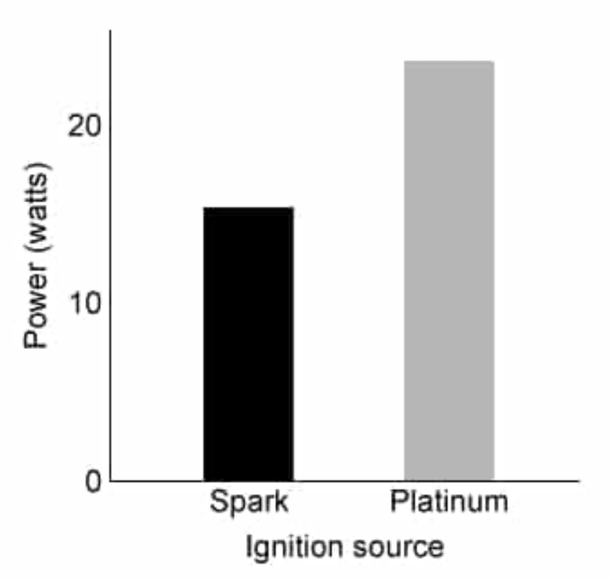
A group of researchers designed two prototype internal combustion engines, each of which could inject a hydrogen/oxygen mixture into a 5-mL combustion chamber at a total pressure of 1 atm and could repeat this process for several cycles. In one engine design, a spark was generated to initiate combustion as hydrogen and oxygen were injected. In another method, the combustion chamber was coated with a small amount of platinum powder. Platinum dramatically increases the reaction rate at room temperature without the need for a spark. The platinum itself is not altered by the reaction and can be reused. Researchers measured the maximum power output of both engine designs under identical injection and temperature conditions (Figure 1).



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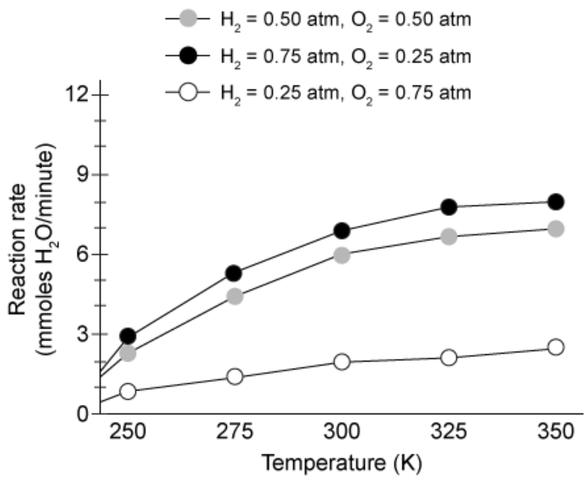


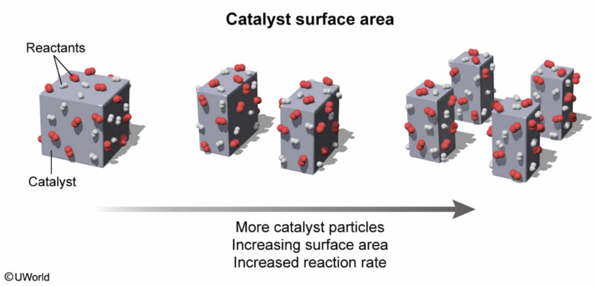
Figure 2 Rate of combustion in platinum-coated engines under various conditions

Which of the following would most likely improve the power output of a platinum-coated combustion chamber? (Note: Assume that the same mass of platinum is used in Choices A, B, and C.)

- A. Coating the chamber with finer platinum powder [75%]
- B. Lining the chamber with larger platinum particles [13%]
- C. Polishing the platinum on the chamber wall to make it smooth [4%]
- D. Decreasing the amount (mass) of platinum to make more room for reactants [6%]

Correct
76% answered correctly

Explanation:



Catalysts increase reaction rates by stabilizing transition states. Surface catalysts do so by **providing a surface** on which more **stable transition states can form**. As such, increased catalyst surface area provides more sites on which transition states can be stabilized, which increases the reaction rate.

The passage states that the power output of an engine is directly related to its rate of combustion. Because platinum functions as a surface catalyst, an increased rate would be achieved by a greater platinum surface area. Breaking a given mass of platinum into smaller particles would create more individual particle surfaces and yield more surface area. Therefore, a **fine platinum powder** would provide more catalytic surfaces and increase the speed of the reaction, resulting in greater power output.

(Choice B) Larger platinum granules made from a given mass would keep more of the platinum atoms in the interior of the granules with fewer atoms exposed to the reactants (ie, less catalytic surface area), resulting in a slower reaction rate with less power output.

(Choice C) Smooth surfaces have less surface area than rough surfaces. As such, a polished platinum wall would result in a slower reaction rate and less power output.

(Choice D) The small amount of platinum in the chamber has no appreciable effect on the amounts of the reactants that can enter the chamber. Decreasing the amount of platinum would decrease the catalytic surface area and result in a slower reaction and less power output.

Educational objective:
Surface catalysts provide a surface on which stable transition states can form. A greater catalytic surface area corresponds to a greater reaction rate.

Time spent:0 sec
Subject:
General Chemistry

QID:401203
System:
Thermodynamics, Kinetics & Gas Laws

Combustion occurs when an oxidation-reduction reaction takes place between a reduced fuel source and an oxidizer, most frequently oxygen. The reaction between hydrogen and oxygen forms water (Reaction 1) and is highly exothermic and thermodynamically favorable. At the completion of hydrogen combustion, essentially all H_2 molecules have been converted to water. Nevertheless, at room temperature and atmospheric pressure, molecular hydrogen and oxygen can coexist quite stably.

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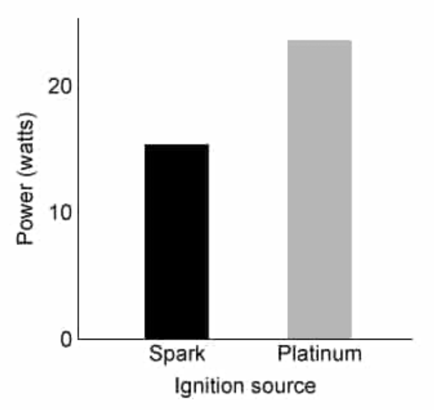

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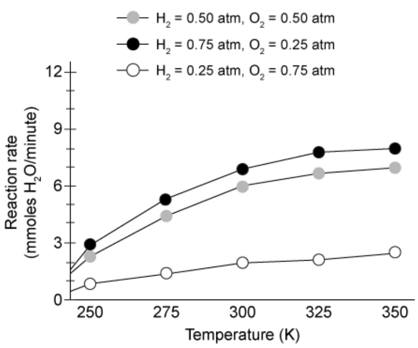


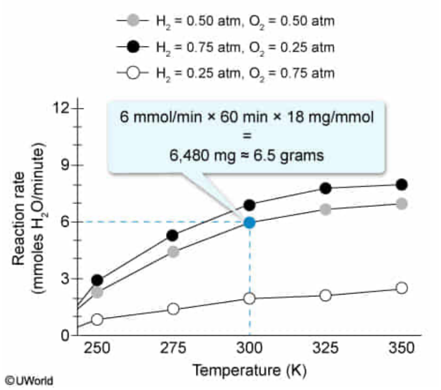
Figure 2 Rate of combustion in platinum-coated engines under various conditions

If the platinum-based engine is run continuously at 300 K for 1 hour with 0.50 atm of H_2 and 0.50 atm of O_2 injected for each reaction, approximately what mass of water will be produced?

- A. 0.36 g [11%]
- ✔ B. 6.5 g [71%]
- C. 360 g [12%]
- D. 6,500 g [4%]

Correct
70% answered correctly

Explanation:



The **amount of product** formed over the course of a reaction can be calculated as the reaction **rate multiplied by the time** the reaction is allowed to occur. Figure 2 shows that at 300 K with 0.50 atm of each reactant injected per reaction (ie, the curve denoted by the gray data points), the platinum-based engine produces water at a rate of 6 mmol/min. If the reaction proceeds at this rate for 1 hour (ie, 60 min), the reaction will produce:

$$6.0\ \frac{\text{mmol H}_2\text{O}}{\cancel{\text{min}}} \times 60\ \cancel{\text{min}} = 360\ \text{mmol H}_2\text{O}$$

To determine the mass of water produced, the number of moles must be multiplied by the molar mass of water (ie, 18 g/mol, or 18 mg/mmol):

$$360\ \cancel{\text{mmol H}_2\text{O}} \times \frac{18\ \text{mg H}_2\text{O}}{\cancel{\text{mmol H}_2\text{O}}} = 6{,}480\ \text{mg H}_2\text{O}$$

Converting the mass to grams gives:

$$6{,}480\ \cancel{\text{mg}}\ \text{H}_2\text{O} \times \frac{1\ \text{g}}{1{,}000\ \cancel{\text{mg}}} = 6.48\ \text{g} \approx 6.5\ \text{g H}_2\text{O}$$

(Choices A and C) The reaction produces 0.36 mol, or 360 mmol, of water in 1 hour. The question asks for the *mass* of this water in grams.

(Choice D) This value is the approximate mass of water produced in milligrams, not grams.

Educational objective:
Given a constant reaction rate, the amount of product formed during a reaction can be calculated as the rate multiplied by the time in which the reaction proceeds. Units may need to be converted between mass and moles.

Time spent:0 sec
Subject:
General Chemistry

QID:401204
System:
Thermodynamics, Kinetics & Gas Laws

Combustion occurs when an oxidation-reduction reaction takes place between a reduced fuel source and an oxidizer, most frequently oxygen. The reaction between hydrogen and oxygen forms water (Reaction 1) and is highly exothermic and thermodynamically favorable. At the completion of hydrogen combustion, essentially all H_2 molecules have been converted to water. Nevertheless, at room temperature and atmospheric pressure, molecular hydrogen and oxygen can coexist quite stably.

$$2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(g)$$

Reaction 1

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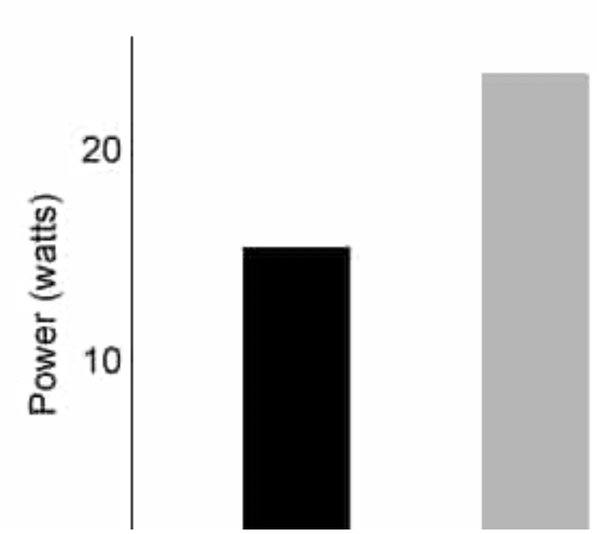

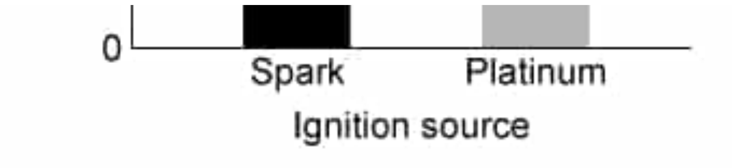

Figure 1 Relative power outputs of hydrogen engines with different ignition sources at 298 K

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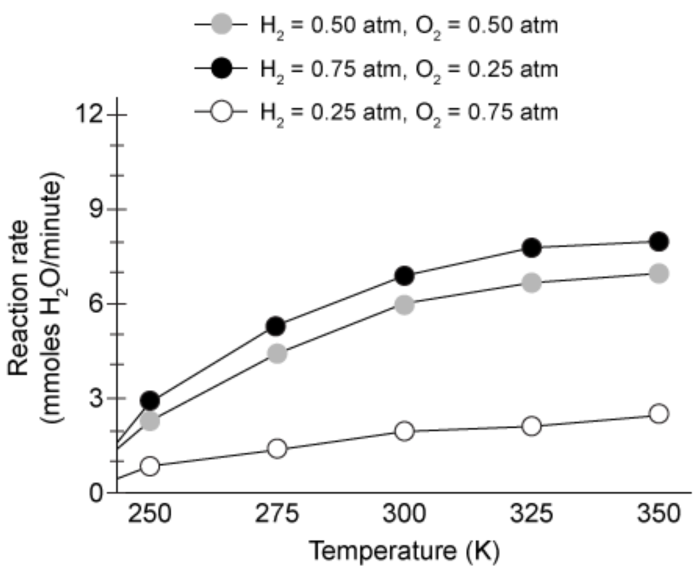


Figure 2 Rate of combustion in platinum-coated engines under various conditions

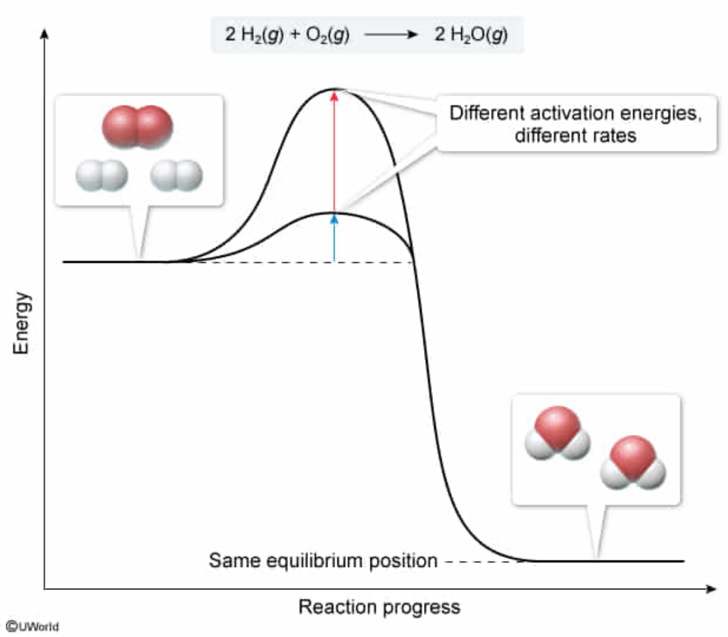
At a given temperature and injection composition, if researchers allowed each engine to carry out only one complete combustion reaction, which engine would produce more water?

- ○ A. The spark engine would produce more water because it has a lower power output. [2%]
- ○ B. The platinum engine would produce more water because it has a higher rate of combustion. [41%]
- ✔ ● C. Both engines would produce the same amount of water because the equilibrium constant is the same for both. [51%]

◯ D. It cannot be determined which engine would produce more water with the available information. [3%]

Correct
51% answered correctly

Explanation:



The amount of product formed in a reaction depends on the equilibrium constant of the reaction. This **constant is not changed** by the presence or absence of a catalyst, which affects only the rate of the reaction.

The passage states that the combustion of hydrogen is thermodynamically favorable with essentially all H_2 molecules being converted to water (ie, at the equilibrium position, all available reactants are consumed). Platinum acts as a catalyst and affects the rate but not the equilibrium position of the reaction. Therefore, both engines would produce the same amount of water in one cycle.

(Choices A and B) Figure 1 indicates that the platinum engine produces more power than the spark engine, suggesting that it consumes the reactants at a higher rate. However, this difference indicates only that the platinum engine completes each cycle more quickly; it does not indicate a change in the equilibrium position or in the amount of water produced once the reaction is complete.

(Choice D) As long as the temperature and initial concentrations of two reactions are identical, they will reach the same equilibrium position regardless of whether a catalyst is present or not. The question states that temperature and injection conditions are identical, so sufficient information is provided to draw this conclusion.

Educational objective:
The amount of product formed in a reaction is dependent on the equilibrium constant of the reaction. This constant is not changed by a catalyst, which only changes the rate at which equilibrium is achieved.

Time spent:0 sec	**QID**:401205
Subject:	**System:**
General Chemistry	Thermodynamics, Kinetics & Gas Laws